

1) Given a micrometer adjustment of

$$\Delta d_{mm} = 1.59 \pm 0.005 \text{ mm}$$

The 5:1 reduction ratio of the lever arm means that the actual distance moved by the mirror is

$$\Delta d = \frac{\Delta d_{mm}}{5} = 318 \pm 1 \text{ } \mu\text{m}$$

Since 4 contrast cycles were observed,

$$\Delta d = 79.5 \pm 0.25 \text{ } \mu\text{m} \text{ (per cycle)}$$

Given spectral line wavelengths 576.961 nm and 579.067 nm,

$$\lambda_{avg} = 578.014 \text{ nm}$$

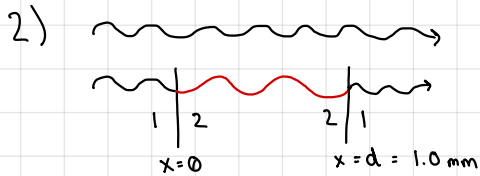
also, this gives an expected value on $\Delta\lambda$,

$$\langle \Delta\lambda \rangle = 2.106 \text{ nm}$$

The result and it's error can be obtained:

$$\begin{aligned} \overline{\Delta\lambda} &= \frac{\lambda_{avg}^2}{2\Delta d} & \sigma_{\Delta\lambda}^2 &= \left(\frac{d(\Delta\lambda)}{d(\Delta d)} \right)^2 \sigma_{\Delta d}^2 & \frac{d(\Delta\lambda)}{d(\Delta d)} &= - \frac{\lambda_{avg}^2}{2(\Delta d)^2} \\ \Delta\lambda &= \frac{(578.014 \text{ nm})^2}{2(79.5 \times 10^3 \text{ nm})} & \sigma_{\Delta\lambda} &= \frac{\lambda_{avg}^2}{2(\Delta d)^2} \sigma_{\Delta d} = \frac{(578.014 \text{ nm})^2}{2(79.5 \times 10^3 \text{ nm})^2} (250 \text{ nm}) = 0.007 \text{ nm} \end{aligned}$$

$$\Delta\lambda = 2.101 \pm 0.007 \text{ nm} \rightarrow \text{The expected result is within } \pm 1\sigma!$$



$$c = \lambda f$$

$$n = \frac{c}{v} \rightarrow n_2 = 1.481, n_1 \approx 1.0$$

$$\frac{n_2}{n_1} = n_2 = \frac{c}{v_2}$$

$$v_1 = c = \lambda_1 f$$

$$v_2 = \lambda_2 f \quad \frac{v_2}{\lambda_2} = \frac{c}{\lambda_1} \rightarrow \frac{c}{v_2} = \frac{\lambda_1}{\lambda_2} = n_2$$

In the glass medium,

$$\lambda_2 = \frac{\lambda_1}{n_2} = 632.8 \text{ nm} / 1.481 = 427.3 \text{ nm}$$

Over a distance $d = 1.0 \text{ mm} = 1.0 \times 10^6 \text{ nm}$, the number of cycles for each wave is given by

$$N_i = \frac{d}{\lambda_i}$$

The phase difference is given in radians by

$$\Delta\phi = 2\pi [(N_2 - N_1) \% 1.0]$$

$$(N_2 - N_1) = \frac{d}{\lambda_2} - \frac{d}{\lambda_1} = \frac{dn_2}{\lambda_1} - \frac{d}{\lambda_1} = \frac{d(n_2 - 1)}{\lambda_1} = \frac{(1.0 \text{ mm})(0.481)}{632.8 \text{ nm}} = 760.114$$

Such that

$$\Delta\phi = 2\pi (0.114) = 0.228\pi \approx \frac{\pi}{4} \text{ radians}$$